AI Use Declaration

ABU-SSO: Unified Single Sign-On & Digital Campus Wallet

Course: COEN838 — Advanced Software Engineering
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Matric No: P25EGCP9003
Date: October 2026

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1. Purpose of This Declaration

The COEN838 exam instructions (item F) require students to "*ensure responsible and ethical use of AI.*" This document fulfils that requirement by transparently disclosing:

1. **Where** AI assistance was used in this project.
2. **How** it was used (as a learning aid, not as a substitute for engineering judgement).
3. **What was NOT delegated** to AI.
4. **How** the final work reflects the candidate's own understanding.

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2. AI Tool Used

Tool	Version / Access	Purpose
DeepSeek	Web interface, October 2026	Conversational coding tutor and design reviewer

No other AI tools were used. No code generation services (e.g., GitHub Copilot, ChatGPT, Cursor) were used to author source files directly.

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3. Nature of Assistance Provided

AI was used in the following **supporting** capacities:

Area	Assistance Type	Candidate's Responsibility
Learning environment setup	Step-by-step guidance on installing Python, Conda, Git, VS Code on Windows	Candidate executed every command and verified each result
Code scaffolding	Example patterns for FastAPI routers, SQLAlchemy models, Pydantic schemas	Candidate reviewed, understood, and where necessary modified each snippet
Debugging	Explaining error messages (import errors, encoding issues, path issues)	Candidate applied fixes and confirmed behaviour

Area	Assistance Type	Candidate's Responsibility
Documentation drafting	Structuring SRS, RTM, Design, Security, and Technical Report sections	Candidate filled in project-specific details and validated factual accuracy
Test case generation	Suggesting edge cases for pytest	Candidate ran tests and inspected failures
Architecture review	Discussing trade-offs (monolith vs microservices, SQLite vs PostgreSQL, JWT vs sessions)	Candidate made the final design decisions

4. What Was NOT Delegated to AI

The following engineering decisions and artefacts were **the candidate's own work**:

1. **Requirements interpretation** — the decision to answer Question 7 (ABU-SSO) and the scope boundary drawn for the prototype.
2. **Design choices** — selecting a modular monolith with clear growth path, choosing JWT + bcrypt over sessions, choosing lazy wallet creation.
3. **Testing strategy** — the decision to use three test layers (unit, service, integration) with a temp database, and to run Bandit SAST.
4. **Deployment** — the decision to split hosting (Render for app, Neon for DB) after discovering Render's free-tier DB expiry.
5. **Written narrative** — the analysis in the Technical Report, Risk Register, and Maintenance Plan reflects the candidate's own understanding of Sommerville and Pressman course material.
6. **Verification** — every code path was executed, every test was run, every deployment was manually verified in a browser.

5. Ethical Use Principles Applied

The following principles guided AI use in this project:

Principle	How it was upheld
Transparency	Every AI interaction is disclosed in this document
Accountability	The candidate takes full responsibility for the correctness, security, and originality of the final submission
Understanding over copying	AI output was treated as a starting point for learning, never as a substitute for comprehension
Verification	Every AI-suggested fix or pattern was tested locally before being accepted into the codebase
Academic integrity	No AI was used to write exam-critical narrative without candidate review; no AI was used during timed exams

Principle	How it was upheld
Attribution	Where AI assistance shaped a design (e.g., the RBAC pattern), the choice was justified on its engineering merits, not on AI authority

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6. Reflection on AI-Assisted Learning

Working with an AI tutor accelerated my learning of:

- FastAPI dependency injection and router patterns
- SQLAlchemy ORM design and session lifecycle
- JWT issuance, verification, and expiry handling
- Test isolation using a temporary database
- Cloud deployment (Render + Neon) and separation of compute vs storage

It did **not** replace the need to:

- Read error messages carefully
- Understand *why* a fix works, not just *that* it works
- Design the system to solve the *stated* problem, not merely to compile
- Write honest documentation that reflects what was actually built

This project demonstrates that AI can be a **legitimate productivity multiplier** when used with discipline, verification, and disclosure.

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7. Declaration

I, **Idris Muhammad Abubakar (P25EGCP9003)**, declare that:

1. The AI use described above is complete and honest.
2. The final software, documentation, and design decisions are my own responsibility.
3. I have verified every component of the submitted system and can defend it orally.
4. This declaration is made in accordance with COEN838 exam instruction F.

Signature: _____
Date: October 2026

End of AI Use Declaration.